

Nicola Dell'Avvocato

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PROFILE

Aerospace Engineering graduate student oriented towards space missions, on-board sensing and aerospace defence. Ambitious, determined and used to leading multidisciplinary teams towards results that are measured rather than described. Seeking a career in the international space and defence industry.

EDUCATION

MSc in Space Engineering Sep 2026 – expected Jul 2028

Politecnico di Torino | Turin, Italy

BSc in Aerospace Engineering Sep 2023 – Jul 2026

Politecnico di Torino | Turin, Italy

- Final grade: 110/110 cum Laude.

Erasmus+ exchange semester Feb 2026 – Jun 2026

Universidade da Coruña | Ferrol, Spain

Merit-based University College — Supplementary Training Programme Sep 2025 – Jul 2028

Collegio Universitario di Torino Renato Einaudi | Turin, Italy

Scientific high school diploma — Cambridge International Curriculum Sep 2018 – Jul 2023

Liceo Scientifico Federico II di Svevia | Altamura, Italy

- Final grade: 100/100.

RESEARCH & PROJECTS

Embedded FBG Sensors for SHM of CFRP Sandwich Structures (BSc thesis) Jun 2026

Politecnico di Torino | Turin, Italy

- Embedded a Fibre Bragg Grating optical fibre directly between the carbon plies of a CFRP sample.
- Built an Abaqus/CAE finite element model of the panel to predict its natural frequencies and strain mode shapes.
- Ran an impact-hammer test campaign with an optical interrogator and processed the strain signals through FFT in MATLAB.

Dynamic Characterisation of 3D-Printed Metamaterials for Aeroelastic Applications Feb 2026 – May 2026

Universidade da Coruña | Ferrol, Spain

- Characterised additively manufactured metamaterial specimens on a dynamic mechanical analysis machine, measuring their damping response as a function of frequency.

INTRAPRENDENTI — Smart Infrastructures Sep 2024 – Jul 2025

Politecnico di Torino | innovation and entrepreneurship programme

- Worked in a multidisciplinary team (engineering, architecture, design) on a smart-infrastructure challenge.

TECHNICAL SKILLS

Simulation & CAD: Abaqus/CAE, finite element analysis (FEA), modal analysis, SolidWorks, CAD/CAM software, 3D modelling

Programming & data: MATLAB, Simulink, Python (libraries: basic knowledge of RocketPy), digital signal processing (DSP), FFT and spectral analysis, data analysis, LaTeX, Microsoft Office (Excel, Word, PowerPoint)

Experimental & laboratory: CFRP composite lay-up and manufacturing, Fibre Bragg Grating (FBG) optical sensors, optical interrogators, impact-hammer modal testing, dynamic mechanical analysis (DMA), strain measurement, structural health monitoring, test-data post-processing

Engineering domains: space systems, composite and sandwich structures, aerodynamics, aeroelasticity, metamaterials and additive manufacturing

Transferable: team leadership and coordination, project management, multidisciplinary teamwork, analytical problem solving, critical thinking, interpersonal communication, community development, work in international environments

Areas of interest: space and military defence, dual-use and UAV technologies, optical fibre sensors, Earth observation, AI applied to aerospace, innovation and entrepreneurship

EXPERIENCE & ACTIVITIES

Staff

Sep 2026 – present

Wave by Vento (formerly Italian Tech Week) | Turin, Italy

- Part of the organising staff of one of Europe's leading technology conferences, which brings together founders, investors and technology leaders.

Head Lifeguard

Jun 2023 – Sep 2023

Tenuta Giannini | Sannicandro di Bari, Italy

- Responsible for pool safety, coordinating the lifeguard team and enforcing safety procedures with the public.

LANGUAGES

Italian: native

English: professional working proficiency

Spanish: professional working proficiency

CERTIFICATIONS & HONOURS

- **Basic Life Support & Defibrillation (BLSD/AED)** — Croce Rossa Italiana.
- **Lifeguard Certification** — Federazione Italiana Nuoto (FIN).
- **Cambridge International Curriculum IGCSE** (*A** - Chemistry, *A** - Latin) · **B2 First** (English), Cambridge International Education.
- **Member of Collegio Universitario di Torino Renato Einaudi** — Merit-based residential college.
- **B2 DELE** (Spanish), Universidade da Coruña
- **INTRAPRENDENTI** — Politecnico di Torino programme for innovative engineers, architects, planners and designers.